

**Arespectfulness:
Invisible Inheritance and the Stack of Solved
Impossibilities**

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Abstract

Modern agents act atop layered inheritances they did not build and cannot fully perceive: the oxygenated atmosphere, the mitochondrial machinery that powers a thought, controlled fire, written memory, the compiler that turns a line of code into machine behavior. This paper develops a concept for the characteristic failure these inheritances produce. Arespect is not disrespect, which perceives a value and diminishes it, and not ingratitude, which is a withheld feeling toward a recognized debt. Arespect is the failure to perceive the depth, cost, and difficulty of an enabling condition at all, the affective and cognitive blankness that settles once a hard-won achievement stabilizes into a reliable interface. Its mechanism is stack blindness: a functioning layer withdraws from attention because we act through it rather than toward it, and it returns to view only when it breaks. The argument has two parts. The first distinguishes arepect from a family of neighboring concepts (disenchantment, reification, alienation, inattentional blindness, the withdrawal of the ready-to-hand, infrastructural invisibility, black-boxing, tacit knowing) and shows that each names either the mechanism by which a stack recedes or the content that recedes, while none names the dispositional failure of regard that the recession makes possible. The second traces the stack itself across scales, from the geological inheritance of breathable oxygen and the cellular inheritance of mitochondrial energy through fire, writing, and software, holding every empirical claim to a strict discipline that separates established science from plausible inference, live controversy, and speculation. The cure proposed is not gratitude but depth perception, including honest perception of how much of the depth remains genuinely unknown. Civilization is read, in the end, as accumulated impossibility rendered usable, and arepect as the price of the rendering.

1. The Floor Was a Mountain

It is very hard to feel how hard fire was. The hominin who first kept an ember alive across a cold night was holding open a door that had taken a chemistry, a meteorology, and a social discipline to find, and almost everything downstream of that night, cooked food, warmth, light, smelting, ceramics, glass, depends on a competence that now hides inside a disposable lighter. The same compression runs through every later layer. A person who has written a compiler by hand, or laid out high-level code that a machine will translate without complaint, knows that the casual command to build a program rests on parsing, type theory, register allocation, instruction sets, and decades of accumulated failure, and knows equally that the next user will type the command and feel none of it. The difficulty does not disappear; the cost is paid once, packaged, and inherited.

This is the ordinary condition of an agent living after accumulation. We operate the top of a very tall stack and experience that top as plain reality. Breath is just breath, text is just text, a cloud deployment is just a push to production. Underneath each “just” sits a buried achievement that was once heroic, rare, expert, dangerous, or very nearly impossible, and the burying is not an accident or a decadence but the way leverage works. A layer earns the right to be forgotten by becoming reliable, and reliability is precisely the property that removes a thing from attention.

The attitude this produces toward the stack is the subject of this paper, and it needs a name that the

available words do not supply. The attitude is not contempt. Almost no one despises the grid, or the alphabet, or oxidative phosphorylation. It is closer to a kind of perceptual blankness, an inability to feel the weight of what one stands on, and the right word for it is *arespect*: the absence of felt regard for inherited difficulty, arising because the difficulty is no longer perceived. The condition is *arespectfulness*; its mechanism, the way working layers drop out of view, is *stack blindness*. The thesis defended here is that *arespect* is a distinct and consequential failure of perception with its own structure, that the structure recurs identically across biological, technical, and cultural scales, and that the failure is built into the very success of the things that occasion it, so that respect, the recovery of depth perception, is an achievement won against a structural tendency rather than a default that mere good will would supply.

2. *Arespect*: A Failure of Regard

Three attitudes toward an enabling condition can be held apart cleanly. Respect perceives the value, cost, and difficulty of the thing and accords it regard. Disrespect perceives the value and actively diminishes it; the despiser has seen the cathedral and called it a shed. *Arespect* does neither, because it has not registered that there was anything to see. The line that matters is the second one. Disrespect is a failure of valuation downstream of perception, and is therefore answerable to argument, since one can be shown that the diminished thing was worth more than one said. *Arespect* is a failure upstream of valuation, a non-perception, and is answerable only to a re-education of attention, since one cannot revalue what one has not noticed. The word itself carries the prescription. To respect, from *respicere*, is to look again. *Arespect* is the condition of never having looked twice, of treating a standing miracle as furniture.

Three properties make the concept precise rather than merely evocative. First, *arespect* is dispositional, a settled stance rather than a momentary lapse; the point is not that one failed to notice the grid this morning but that one's whole relation to the grid is one of unfelt reliance. Second, it is evaluative without being moralizing; to call a stance *arespectful* is to say it fails to track a real depth, which is a claim about accuracy of perception before it is a claim about blame, and an agent can be fully innocent and thoroughly *arespectful* at once. Third, its object is specifically inherited difficulty, the sedimented labor and solved problems that a layer contains, rather than the layer's surface utility, which *arespect* perceives perfectly well. The *arespectful* user of a bridge sees the bridge and crosses it; what goes unseen is the bridge as a settled achievement, the metallurgy and the load calculation and the maintenance that hold it up.

The concept earns its place only against the neighbors it might collapse into, and the family is large. Each neighbor names something real and adjacent, and each names either the mechanism by which a layer recedes from view or the content that recedes, while leaving unnamed the dispositional failure of regard that is *arespect* proper.

Concept	What it names	How arepect differs
Disenchantment (Weber, 1946)	The cultural draining of mystery from a world made calculable	Removes wonder from the knowable; arepect removes regard for difficulty, and can coexist with full re-enchantment
Reification (Lukács, 1971)	Human relations appearing as relations between things	Misreads the human as thing; arepect fails to perceive the difficulty inside the thing, and persists outside any commodity form
Alienation (Marx, 1959)	The producer's estrangement from their own labor and its product	Concerns the producer's own activity; arepect concerns the beneficiary's relation to others' sedimented labor
Inattentional blindness (Mack & Rock, 1998; Simons & Chabris, 1999)	Failure to perceive an unattended stimulus under load	A momentary, capacity-limited lapse; arepect is a settled disposition that survives full attention
Change blindness (Rensink et al., 1997)	Failure to detect a change across views	A detection failure; arepect is a valuation failure, intact perception with absent regard
Misplaced concreteness (Whitehead, 1925)	Mistaking an abstraction for the concrete reality	Takes the abstract as real; arepect flattens the concrete depth of the real into a frictionless given, the inverse error
Ready-to-hand withdrawal (Heidegger, 1962)	The working tool receding from notice in skilled use	Names the phenomenological mechanism of recession; arepect is the stance that hardens once the receded is taken for empty
Infrastructural invisibility (Star, 1999)	The structural property by which working infrastructure goes unseen	A property of systems; arepect relocates the phenomenon in the disposition of the perceiver

Concept	What it names	How arepect differs
Tacit knowing (Polanyi, 1966)	The structure of attending from subsidiary particulars to a focal whole	Describes a necessary and non-pathological reliance; arepect is the pathology of mistaking the subsidiary for the nonexistent

The table is also the argument. Reading down the middle column, every entry describes how or why a layer drops out of awareness, or what it is that drops out, and the recession each describes is in general necessary, functional, and often virtuous: one could not act at all while focally attending to every condition of acting. Arepect is what the right-hand column isolates, the failure that becomes possible when a necessary recession is mistaken for a metaphysical fact, when “I need not attend to this in order to use it” curdles into “there is nothing here worth my regard.” The neologism is earned precisely because the failure is none of the neutral mechanisms that make it tempting.

3. Why the Stack Withdraws

The mechanism that makes arepect possible is independently well described, from several directions, and the convergence is itself instructive. Heidegger’s analysis of equipment in *Being and Time* gives the phenomenological root. In skilled use the tool is *zuhanden*, ready-to-hand, and its readiness consists in a kind of self-effacement: the hammer is not an object of notice but the medium through which attention reaches the work, and the more competent the use, the more completely the hammer withdraws (Heidegger, 1962). The tool returns to view only when it fails, when the head flies off or the nail bends, at which point it shows up as a mere present-at-hand thing with properties, conspicuous exactly in proportion to its uselessness. Functioning is invisibility; breakdown is the return of the seen.

Susan Leigh Star generalized the same structure from tools to infrastructure and gave it a sociology. Working infrastructure, by its nature, is embedded, learned as part of membership, built on an installed base, and transparent in use, and from these properties follows the one that matters here: it becomes visible upon breakdown (Star, 1999; Star & Ruhleder, 1996). The server is down, the bridge washes out, the power fails, and the normally unseen substrate snaps into view, often along with the maintenance arrangements that had been holding it silently in place (Bowker & Star, 1999). Invisibility, on this account, is not a defect of infrastructure but a condition of its working; a road one had to consciously assemble for each journey would not be a road.

Bruno Latour named the epistemic version of the same paradox with a precision that is hard to improve. Black-boxing, he wrote, is the way scientific and technical work is made invisible by its own success: when a machine runs efficiently, when a matter of fact is settled, one need focus only on its inputs and outputs and not on its internal complexity, so that, paradoxically, the more science and technology succeed, the more opaque and obscure they become (Latour, 1999). The

clause to hold onto is the last one. Opacity here is not a residue of failure to be cleared away by better explanation but a product manufactured by success, increasing with it. A technology that everyone understood in detail would be one that had not yet receded enough to be useful.

Michael Polanyi supplies the cognitive grammar that ties these together and, crucially, marks the boundary that *arespect* crosses. In any skilled performance, Polanyi observed, we attend from a set of subsidiary particulars to a focal target: the pianist attends from the keys to the music, the reader from the letters to the meaning, and to switch focal attention onto the subsidiaries, to stare at the individual keys mid-phrase, is to destroy the performance (Polanyi, 1966). We rely on the stack by dwelling in it, by seeing through it, and this attending-from is not a vice but the very form of competence. The decisive point follows immediately. Attending-from is structurally necessary and entirely healthy; the pathology is not the failure to attend to the subsidiary but the slide from “I rightly attend from this” to “there is nothing here to attend to.” *Arespect* is that slide, and only that slide. It is what happens when the functional and proper recession of a working layer is taken as evidence that the layer has no depth, no cost, and no claim.

Two further sources supply not the mechanism of recession but its content, what it is that recedes. Gilbert Simondon argued that culture relates to technical objects through a structural blindness to their internal genesis, treating them either as mere assemblages of utility or as alien threats, and in both cases failing to grasp that a technical object is the outcome of a long individuation, a convergence toward internal coherence that he called concretization, carrying its formative history folded inside it (Simondon, 2017). What recedes, on this account, is exactly the genesis, the sedimented sequence of solved problems that makes the object what it is. Bernard Stiegler pressed the temporal point further: the human is constituted by the exteriorization of memory and experience into technical supports, an inheritance he called tertiary retention, and this inheritance reaches each newcomer already complete, with the labor of its formation amputated, so that forgetting is built into the very structure by which technics is transmitted (Stiegler, 1998). The stack, in Stiegler’s terms, is inherited externalized memory, and it arrives stripped of the record of its own difficulty. This is why *arespect* is the default and *respect* the achievement. The mechanisms of recession are universal and the content that recedes is, by the manner of its transmission, pre-erased.

4. Breath as Inherited Geology

The stack does not begin with tools. The deepest layers an agent inherits are biological and geological, and they exhibit the same structure of buried difficulty with even less hope of being felt, because they were laid down before anything existed that could feel. The clearest case is the air. The oxygen that a reader is now consuming, at a rate the body never consciously authorizes, is not a neutral background condition of life on Earth but the accumulated residue of a planetary-scale biological transformation, and treating it as simply given is *arespect* toward a geological inheritance.

For most of the Archean the atmosphere was effectively anoxic, holding free oxygen at something below a hundred-thousandth of present levels. The permanent rise of atmospheric oxygen, the Great Oxidation Event, is conventionally placed in the early Paleoproterozoic, with its onset brack-

eted around 2.4 to 2.3 billion years ago and the broader transition extending later still, and its primary diagnostic is the disappearance from the rock record of a mass-independent sulfur-isotope signal that can only survive under an oxygen-poor sky (Lyons et al., 2014; Bekker et al., 2004). The agent of the change was oxygenic photosynthesis in cyanobacteria, and a feature of the timeline carries the argument: the biological capacity to produce oxygen appears to predate its permanent atmospheric accumulation by a long interval, on the order of hundreds of millions of years, because the early oxygen was consumed by reduced sinks in the oceans and crust before it could build up (Lyons et al., 2014). The most recent high-resolution constraints place the coupled oxygenation of atmosphere and ocean closer to 2.3 billion years ago and emphasize how protracted and halting the transition was (Poulton et al., 2024). The exact onset remains genuinely contested across a wide window, and the breath should be dated as the outcome of a slow planetary process rather than a single event.

The transformation was, from the standpoint of the life that preceded it, a chemical threat. Molecular oxygen and its partially reduced derivatives, the reactive oxygen species, oxidize iron-sulfur clusters, mononuclear iron centers, and DNA, and the mechanism of this damage is understood in detail and is textbook-solid (Imlay, 2003; Imlay, 2013). The response was the evolution of antioxidant defenses, the superoxide dismutases, catalases, and peroxidases that aerobic life still runs continuously, so that the very inheritance which now sustains respiration first had to be survived. Here the discipline of the argument matters, because the popular framing overreaches. There is no direct evidence for a discrete mass extinction of anaerobic life, an “oxygen catastrophe” in the sense of a documented die-off, and the microbial record of the period is far too sparse to support such a claim; what is defensible is that rising oxygen imposed a strong selective pressure, drove the evolution of antioxidant machinery, and confined obligate anaerobes to the anoxic refugia where they still thrive (Sessions et al., 2009). Oxygen was simultaneously a poison to be defended against and an energy source of extraordinary richness, the basis of an aerobic metabolism many times more efficient than fermentation, and both facts hold at once.

The chronology reaches back further than the air. The last universal common ancestor, the cellular population from which all known life descends, is reconstructed as an anaerobe, and this much is bedrock, over-determined by the simple fact that the early Earth held no free oxygen to be used. The most influential reconstruction goes considerably further, inferring from a set of genes traceable to that node a hydrogen-dependent, carbon-dioxide-fixing metabolism rooted in a hydrothermal setting (Weiss et al., 2016), and here the discipline reasserts itself, because that richer portrait is a strong inference rather than a settled fact, vulnerable to undetected lateral gene transfer and challenged by later work that revises the organism’s age and gene complement (Moody et al., 2024). The safe claim is the spare one: the ancestor was anaerobic, and the oxygenated world that its descendants now treat as the obvious home of life would have been biochemically alien, and very likely hostile, to it. The breath is not immediate but inherited planetary biochemistry, paid for in deep time by organisms that could not have meant us any benefit.

5. The Engine in the Cell

Between the oxygenated air and the thought that contemplates it sits a third inherited layer, the one that converts the first into the second. Aerobic eukaryotic life runs on mitochondria, and their origin is one of the few deep-evolutionary facts secure enough to bear weight. Mitochondria descend from a free-living alphaproteobacterium taken up by an archaeal host in a single endosymbiotic event, a thesis revived by Lynn Margulis in the nineteen-sixties and now placed beyond serious doubt by phylogenomics (Margulis, 1967; Muñoz-Gómez et al., 2022). What remains debated is the order of events, whether the acquisition of the bacterium triggered the origin of the complex cell or merely punctuated an already-complexifying lineage, a question on which the hydrogen hypothesis, the discovery of the Asgard archaea as the closest prokaryotic relatives of eukaryotes, and the cultivation of a syntrophic Asgard archaeon all bear without settling it (Martin & Müller, 1998; Spang et al., 2015; Imachi et al., 2020). The inheritance is certain; its founding moment is still under reconstruction.

The mitochondrion converts energy by a mechanism whose discovery overturned a decade of assumptions and which is now as solid as anything in cell biology. Peter Mitchell's chemiosmotic hypothesis held that the respiratory chain pumps protons across the inner mitochondrial membrane to build an electrochemical gradient, a proton-motive force, and that this gradient drives the synthesis of ATP, the cell's energy currency (Mitchell, 1961). The inner membrane is folded into cristae, and the folding is functional rather than decorative: cristae shape, set by a defined protein machinery and by rows of ATP-synthase dimers that bend the membrane along their ridges, organizes the respiratory chain and tunes the efficiency of energy conversion (Cogliati et al., 2013; Cogliati et al., 2016). A great deal of what makes a thought metabolically possible is settled here, in the geometry of a membrane most thinkers will never picture.

A famous and elegant argument tries to make the mitochondrion the cause of complex life itself, and the discipline of this paper requires that it be reported as the live controversy it is. Lane and Martin proposed that prokaryotes face a bioenergetic ceiling, because they generate ATP across their outer surface and scaling up yields diminishing returns, and that mitochondria broke the ceiling by internalizing bioenergetic membranes with their own local genomes, supplying orders of magnitude more energy per gene and thereby permitting the expansion of genome and complexity that underlies eukaryotic life (Lane & Martin, 2010). The claim is influential and the mechanism is plausible, but the central quantitative step has been directly challenged: Lynch and Marinov recomputed the energetic cost of a gene relative to cell volume across both domains and concluded that mitochondria conferred no decisive energetic advantage, since eukaryotes surrendered surface bioenergetics in the bargain (Lynch & Marinov, 2015), and others have questioned the energy-first framing of eukaryogenesis altogether (Booth & Doolittle, 2015). The proposition that mitochondria made complexity possible is therefore an open hypothesis, and any use of it must be paired with its rebuttal rather than asserted as established.

The careful version of the inheritance, the one the evidence will bear, runs through cognition by way of energy budgets rather than grand causal claims. The brain is metabolically extravagant: roughly two percent of body mass, it accounts for on the order of twenty percent of the body's

resting energy expenditure, and that proportion is remarkably stable across mental effort (Raichle & Gusnard, 2002). The bulk of the spending is not housekeeping but signaling, the electrochemical work of firing action potentials and then restoring the ion gradients that firing dissipates, with detailed budgets attributing the dominant share of grey-matter energy use to action potentials and postsynaptic currents (Attwell & Laughlin, 2001; Harris et al., 2012). That work is performed by ATP-dependent pumps supplied, at the synapse, by mitochondria, which are physically trafficked to presynaptic terminals to deliver ATP and buffer calcium, and whose local energy reserves are required for the protein synthesis that underwrites synaptic plasticity (Devine & Kittler, 2018; Rangaraju et al., 2019). Neurons' high and continuous demand also makes them disproportionately vulnerable when this machinery falters, which is why mitochondrial dysfunction is a recurring feature of neurodegeneration, and why chronic stress registers in mitochondrial state, though the causal direction in disease remains partly open (Picard & McEwen, 2018).

It is therefore defensible, and only this much is defensible, to say that cognition rests on a bioenergetic infrastructure: mitochondrial ATP and calcium handling are enabling conditions of neural computation, such that their failure degrades it. What this does not license is the seductive next step, the claim that the geometry of cristae, or the flux of protons, produces or explains conscious experience. The strongest work that takes brain energetics seriously is explicit that energy is a predisposition for consciousness rather than its correlate, leaving the explanatory burden to mechanistic theories of neural integration (Chen & Zhang, 2021). A particular confusion must be marked and avoided, because it is easy to fall into. The "free energy" of the free-energy principle in theoretical neuroscience is an information-theoretic quantity, a measure of surprise relative to a model, and it has nothing directly to do with the metabolic free energy that mitochondria supply (Friston, 2010); to slide between the two senses is a category error that would dress an equivocation as a discovery. The honest seam is interoception, the brain's representation of the body's own physiological condition, which sits close to the regulation of metabolic state and is a candidate substrate for the felt sense of being a living body (Craig, 2002; Barrett & Simmons, 2015), and which must be handled as the open and careful research frontier it is rather than as a bridge already crossed. The subject who says "I think" does so atop an energetic infrastructure it did not author and cannot suspend, and that sentence is true and worth feeling. The sentence that would have cristae think is not licensed, and the difference between the two is the whole discipline.

6. Fire and the Externalized Mind

Above the biological layers the human stack proper begins, and it repeats the pattern in a register that can at least in principle be reconstructed historically. Fire is the hinge. The control of fire externalized digestion: cooking gelatinizes starch, denatures protein, and kills pathogens, raising the energy a body can extract from food and lowering the cost of getting it, and on the cooking hypothesis this energetic windfall is what permitted the human brain, an organ as metabolically expensive as the preceding section described, to expand while the gut shrank to pay for it (Wrangham, 2009; Wrangham et al., 1999; Aiello & Wheeler, 1995). The hypothesis is powerful and contested at its most load-bearing joint, the timing, and intellectual honesty requires stating the difficulty

rather than smoothing it. Secure evidence of controlled fire is old but isolated, with burned bone and ashed plant material at Wonderwerk Cave dated to around a million years ago and a hearth-bearing site at Gesher Benot Ya'aqov to around seven hundred ninety thousand years (Berna et al., 2012; Goren-Inbar et al., 2004), yet the evidence for *habitual* fire use, fire reliable enough to drive an evolutionary dependency, becomes continuous only much later, perhaps four hundred to three hundred thousand years ago, which strains the causal timeline the cooking hypothesis requires (Roebroeks & Villa, 2011). The strict brain-gut trade-off has likewise been qualified by analyses finding no simple inverse relation between brain and digestive-tissue size across primates (Navarrete et al., 2011). What survives the controversy is the structural claim: a metabolic achievement was externalized into a technique, the technique became a precondition for the kind of animal that uses it, and the precondition is now invisible inside the everyday act of eating a cooked meal.

Writing externalized memory, and did so first as accounting. The earliest writing in the Mesopotamian sequence grew out of a system of clay tokens used to record quantities of goods, the tokens giving way to marks impressed on clay envelopes and then to cuneiform proper, so that the technology now associated with literature and law began as a device for tracking debt and inventory (Schmandt-Besserat, 1992). What writing accomplished, on the strong reading associated with the study of orality and literacy, was to place memory outside the skull, to make thought durable, transmissible across distance and death, and available for the kind of inspection that lists, contracts, and tables permit (Ong, 1982; Goody, 1977; Goody & Watt, 1963). The further claim, that literacy restructured the very form of consciousness, dividing literate from oral cognition along a sharp line, is a genuine live controversy and should be flagged as one, since the so-called Great Divide has been forcefully criticized on the ground that literacy's effects are local to the social practices that carry it rather than uniform consequences of the technology itself (Street, 1984).

The decisive observation about writing is the oldest, and it is a near-perfect early statement of are-spect. In Plato's *Phaedrus*, the god Theuth presents writing to the king as a remedy for memory and forgetting, and the king replies that it is the opposite, a recipe not for memory but for reminding, which will implant forgetfulness in the souls of those who learn it, since they will trust the external marks and cease to remember from within (Plato, 1995). The complaint is itself an instance of the thing this paper describes, a moment of vivid perception of a layer at the instant of its installation, before the recession sets in. Twenty-four centuries later the alphabet is so wholly inherited, so completely a medium attended-from rather than toward, that the reader of this sentence sees through the letters to the meaning without the faintest sense of the abstraction that phonetic writing required, which is exactly the blankness the king foresaw, arrived on schedule.

That no individual could rebuild this stack is not a rhetorical flourish but an empirical finding of the study of cumulative culture. Human capability is the product of a ratchet, a cultural transmission faithful enough to retain improvements and so to accumulate them across generations beyond what any single mind could invent (Tomasello, 1999), and the dependence this creates is total and easily demonstrated by its failures. Well-equipped explorers have starved in environments that sustained the populations native to them, because survival there depended on a large body of

locally accumulated know-how, about which plants must be processed to be safe and how, that the explorers lacked and could not reconstruct from first principles in the time available (Henrich, 2016). The know-how was a stack, inherited by the locals and invisible to them as a stack, and lethal in its absence. Stack blindness is not a failure of intelligence. The explorers were not unintelligent but *arespectful*, in the precise sense of being unable to perceive the depth of the inherited competence whose absence killed them.

7. The Designed Black Box

Computing is where the stack becomes self-conscious, because here the burying of difficulty under interface is not a byproduct of success but an explicit design goal, pursued by name. The history that the casual command to compile a program hides is recent enough to be documented and dramatic enough to make the compression vivid. When Grace Hopper built early compiling systems at the start of the nineteen-fifties, the proposal that a machine could translate human-readable instructions into its own code was met with disbelief, on the assumption that programming was an arithmetic craft a computer could not perform on its own behalf (Hopper, 1952), and when the FORTRAN team delivered a compiler whose output rivaled hand-written assembly later in the decade, it did so against a widespread conviction that automatic translation would necessarily be too inefficient to use (Backus et al., 1957). The conviction was reasonable and wrong, and its defeat installed a layer. Every later programmer inherits the compiler as a given and feels none of the doubt that had to be overcome to build it.

What is distinctive about the technical stack is that its opacity is engineered on purpose, and the engineering has a literature and a vocabulary. Edsger Dijkstra defended abstraction against the charge of vagueness by inverting it: the purpose of abstraction, he argued, is not to be vague but to create a new semantic level at which one can be absolutely precise, a level on which one reasons without descending into the one below (Dijkstra, 1972). David Parnas turned the principle into a discipline of design with the doctrine of information hiding, the deliberate concealment of a module's internal workings behind an interface so that the rest of a system can use it without knowing, or being able to depend on, how it works inside (Parnas, 1972). Information hiding is stack blindness specified as an engineering requirement; the inability to see into the layer is the feature being built. Herbert Simon gave the general systems rationale a generation earlier, observing that complex systems are typically hierarchic and nearly decomposable, organized into layered subsystems whose internal interactions are dense and whose cross-layer interactions are sparse, and illustrating with a parable of two watchmakers in which the one who builds in stable sub-assemblies finishes while the one who must hold the whole mechanism in mind at once never does (Simon, 1962). Layering is how complexity is built at all, and to build in layers is to build in invisibility.

The technical stack therefore mirrors the biological one, with one difference worth stating plainly. The compiler was designed to hide its workings; the mitochondrion was not designed at all. Yet the consequence for the user is identical, a reliable interface attended-from rather than toward, and the technical case has the advantage of exhibiting the breakdown-reveals-the-stack structure in a

controlled form. The practitioner’s law of leaky abstractions records exactly this: all non-trivial abstractions, to some degree, leak, and the substrate they were built to hide reasserts itself at the margins, in the performance cliff or the strange error that forces the programmer to descend a level and confront what the abstraction was holding back (Spolsky, 2002). The leak is the engineered analogue of Heidegger’s broken hammer and Star’s failed server, the moment the black box springs open and its contents become briefly, unwelcomely visible. The same essential difficulty persists beneath the accidental conveniences layered over it, a distinction the discipline itself has long drawn between the complexity intrinsic to a problem and the complexity merely imposed by inadequate tools (Brooks, 1987). Artificial intelligence is the newest layer of this kind, a statistical compression of an enormous quantity of inherited human culture into a callable interface, and it occasions a respect of a particularly pure form, since its users attend to a fluent output while the training corpus, the labor that produced it, and the machinery that compresses it recede completely, a black box whose internal complexity is opaque even to those who build it.

8. What Can and Cannot Be Said

A paper that diagnoses the failure to perceive depth incurs a special obligation to perceive accurately the depth of its own claims, including the parts that are not yet known. The cure for a respect is depth perception, and depth perception includes seeing clearly where the floor of current knowledge actually is rather than papering it over with confident prose. The empirical claims that the argument leans on are not all of equal strength, and honesty requires grading them and marking, for each, the overclaim that the careful version refuses.

Claim used in the argument	Status	Key sources	Overclaim the paper refuses
Breathable oxygen is the accumulated product of cyanobacterial photosynthesis, risen permanently around 2.4 to 2.3 Ga	Established as to cause; the precise onset is contested across a wide window	Lyons et al. (2014); Bekker et al. (2004); Poulton et al. (2024)	That the rise was a single datable event rather than a protracted transition
Rising oxygen chemically threatened anaerobic life and selected for antioxidant defenses	Mechanism established; the scale of any die-off is not	Imlay (2003, 2013); Sessions et al. (2009)	That the event was a documented mass extinction, an “oxygen catastrophe”
LUCA was anaerobic	Bedrock, over-determined by an anoxic early Earth	Weiss et al. (2016); Moody et al. (2024)	That its hydrothermal, hydrogen-dependent metabolism is settled fact

Claim used in the argument	Status	Key sources	Overclaim the paper refuses
Mitochondria arose once by endosymbiosis of an alphaproteobacterium	Established	Margulis (1967); Muñoz-Gómez et al. (2022)	That the early-versus-late timing of the event is resolved
ATP synthesis proceeds by chemiosmosis across folded cristae	Established	Mitchell (1961); Cogliati et al. (2013, 2016)	That cristae geometry has any settled relation to conscious experience
Mitochondria removed a bioenergetic ceiling and so enabled eukaryotic complexity	Live controversy	Lane & Martin (2010) versus Lynch & Marinov (2015); Booth & Doolittle (2015)	That the energy-per-gene thesis is established; it must be paired with its rebuttal
Cognition depends on mitochondrial ATP and calcium handling as enabling conditions	Established as enabling-condition	Attwell & Laughlin (2001); Rangaraju et al. (2019); Devine & Kittler (2018)	That bioenergetics constitutes or produces consciousness
Cooking externalized digestion and helped permit brain expansion	Plausible inference; the timing is contested	Wrangham (2009); Aiello & Wheeler (1995) versus Roebroeks & Villa (2011); Navarrete et al. (2011)	That habitual fire is securely old enough to have driven the expansion
Writing externalized memory and began as accounting	Established	Schmandt-Besserat (1992); Ong (1982)	That literacy uniformly restructures consciousness along a Great Divide

The table is not an apology appended to the argument but a part of it. The structural thesis, that inherited difficulty recedes from perception and that the recession is built into the success of what recedes, does not depend on any single contested empirical claim, because it is exhibited equally by the secure layers and the contested ones. The reader who is unconvinced that mitochondria caused complex life can drop that claim entirely and lose nothing, because the chemiosmotic machinery is no less an unfelt inheritance for the dispute about what it made possible. Calibration of this kind is itself the practice the paper recommends. To perceive the depth of the stack accurately is to perceive, among other things, how much of it remains genuinely open, and a respect that required the depths to be fully known would be a sentimentality rather than a discipline.

9. Objections and Replies

The strongest objections to the concept are worth stating at their full force, because the concept improves under the pressure.

The first objection is that *arespect* is merely ingratitude wearing a coat, a moralizing complaint that people fail to be thankful enough for civilization, dressed up as a perceptual category to disguise its sentimentality. The reply is that the central claim is perceptual before it is evaluative. To call a stance *arespectful* is in the first instance to say that it fails to track a real feature of the world, the sedimented difficulty of an enabling condition, in the same way that a failure to see depth in a visual field is a failure of accuracy rather than of virtue. An agent can be wholly without obligation in the matter and still be *arespectful*, because non-perception is not a withheld payment. Gratitude is a possible response to perceived depth; *arespect* is the prior absence of the perception that would give gratitude or its refusal anything to be about.

The second objection is the most serious, and it comes from Polanyi's own analysis. If competent action requires attending from the stack rather than to it, then the recession of the stack is not a failure but a condition of doing anything at all, and a demand to attend to every inherited layer would be a demand to be paralyzed. This is correct, and the concept is built to concede it without collapse. *Arespect* is not the attending-from, which is necessary and healthy, but the hardening of attending-from into the metaphysical verdict that there is nothing to attend to. The pianist who plays through the keys is not *arespectful* toward the piano; the listener who concludes from the seamless performance that playing is easy, that there was no cathedral of training and tuning and manufacture beneath the music, is. Respect does not require constant focal attention on the subsidiary. It requires only that the functional recession not be mistaken for emptiness, a standing disposition to know that the depth is there even while one rightly sees through it.

The third objection is that the concept licenses a romantic primitivism, a counsel to tear away the interfaces and return to an unmediated appreciation of fire and soil. The reply is that depth perception is the opposite of renunciation. The aim is to inherit the stack and perceive it as inherited, to use the compiler and know what the compiler hides, and Stiegler's analysis makes clear why this is the only coherent option: since the human is constituted by technical inheritance, there is no unmediated ground to return to, and the choice is only between an *arespectful* relation to the inheritance and a respectful one (Stiegler, 1998). Respect is the achievement of holding the depth in view while continuing to depend on the layer, which is harder than either naive use or nostalgic refusal and is the actual recommendation.

The fourth objection is that the biological argument is speculative and that resting a philosophy of perception on contested claims about mitochondria and consciousness is itself a kind of overreach. The reply is the entire eighth section. The argument is held to an enabling-condition claim that is established, the dependence of neural computation on mitochondrial energy, and explicitly refuses the constitutive claim that would make cristae the seat of mind; the contested propositions are flagged as contested and are not load-bearing. A diagnosis of overclaiming that practiced overclaiming would refute itself, and the discipline is therefore not an external constraint on the

argument but an expression of it.

The fifth objection asks why a new word is needed when the phenomenon has been described by Star, Heidegger, Latour, and others already. The reply is the burden of the second and third sections. Those accounts name the mechanism by which working layers recede, or the content that recedes, and they are indispensable and fully credited; what none of them names is the dispositional failure of regard that the recession makes possible, the settled stance that takes the functionally invisible to be actually empty. Invisibility is a property of infrastructure, withdrawal a structure of equipment, black-boxing an economy of attention, and each is in general benign. Arespect is the vice that these benign mechanisms make available, and a concept is warranted exactly when it isolates something the existing vocabulary leaves unmarked.

10. Civilization as Accumulated Impossibility Rendered Usable

The pattern has held at every scale the argument has touched. A difficult achievement, the oxygenation of a planet, the internalization of a bacterial power plant, the control of fire, the externalization of memory, the automatic translation of language into machine behavior, stabilizes into a reliable layer; the layer is attended-from rather than toward; the attending-from, necessary and proper, hardens into the perception that there is nothing beneath; and the achievement is inherited as a frictionless default by agents who feel none of its cost. Civilization, read this way, is accumulated impossibility rendered usable, a tower of solved problems each of which has been buried under the interface that solving it produced, and arespect is the characteristic relation of those who live near the top of the tower to the depth below them.

The relation has a cost that is not merely contemplative, and the maintenance literature locates it. The same blindness that hides the genesis of a layer hides the labor that keeps it standing, so that the people who repair and sustain infrastructure work in a structural invisibility that the cult of innovation deepens, their effort unseen precisely because success is silent and only failure speaks (Russell & Vinsel, 2016; Vinsel & Russell, 2020). A use-centered rather than novelty-centered account of technology makes the scale of the omission plain, since maintenance consumes far more of the world's technological effort than invention does while receiving almost none of its attention (Edgerton, 2007). Arespect toward the stack is therefore continuous with arespect toward the maintainers of the stack, and the recovery of depth perception has a distributive edge: to see the bridge as a sustained achievement is to see the people sustaining it.

What the argument recommends is not gratitude, which can be performed without perception, and not a paralyzing reverence that would have one stare at the keys mid-phrase. It is the cultivated capacity to know that the floor was once a mountain even while walking across it without effort, to attend from the stack and still hold its depth in view, to use the inheritance and perceive it as inheritance. The word for this is the literal sense of respect, *respicere*, to look again, and the looking-again is an achievement against a tendency rather than a sentiment one either has or lacks, because the stack is built, at every layer, to make itself unnecessary to perceive. The first hominin who kept the ember alive could still feel what fire cost. The discipline this paper names is the deliberate recovery, against the whole momentum of inheritance, of something of what that hominin felt:

that the breath is borrowed from cyanobacteria, that the thought is paid for in proton gradients, that the words on this page are an externalized memory the king of Egypt mistrusted, and that beneath every ordinary act lies a buried difficulty that someone, or some lineage, or some planet, once had to overcome.

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